Per-candidate mass uncertainty $\sigma(m_{KK\pi})$ from momentum-calibrated tracks

R(D_s) analysis — working note

3 November 2026

1 The idea

The per-candidate uncertainty $\sigma(m_{KK\pi})$ is computed from what each track's own fit says about its momentum, propagated linearly to the invariant mass, with the resolution added by the momentum calibration folded in. The vertex fit plays no part. The result is a function of reconstructed quantities only, so it is defined identically in data and simulation and can be used to build categories.

1. **Inputs.** For each of K_1, K_2, π take the 3×3 block of the helix covariance in $((q/p), \lambda, \phi)$. The three tracks are uncorrelated.
2. **Propagation.** $\sigma^2(m) = \sum_i J_i\, C_i\, J_i^T$, with $J_i = \partial m / \partial((q/p), \lambda, \phi)_i$ evaluated at the calibrated momenta.
3. **Calibration term.** Add to each track's (q/p) variance the extra relative resolution that the calibration assigns to its (p, p_T, η, q) bin, computed without the random number z.
4. **Same definition everywhere.** Apply steps 1–3 unchanged to data and MC. Never use anything that exists only in MC (z, the realised smear, generator information).

The muon is not part of $m_{KK\pi}$ and never enters. Everything below generalises to m_{KK} or $m_{KK\pi\mu}$ by changing the set of tracks in the sum.

2 Step 1: the inputs

Only the 3×3 momentum block of each track's covariance enters the mass, and the three tracks are independent.

What the 5×5 matrix describes. A CMS track is a helix with five parameters at its reference point (the point of closest approach used by the packed candidate). They are stored, in this order, as `<trk>_cov_<a>_<b>` (upper triangle):

parameter	meaning	unit
q/p	charge over total momentum	GeV^{-1}
λ	dip angle, $\lambda = \pi/2 - \theta$; $\sin\lambda = \tanh\eta$, $\cos\lambda = 1/\cosh\eta$	rad
ϕ	azimuth of the momentum at the reference point	rad
d_{xy}	transverse impact parameter	cm
d_{sz}	longitudinal impact parameter	cm

The momentum depends on the first three only:

$$\vec{p} = p\,(\cos\lambda\cos\phi,\ \cos\lambda\sin\phi,\ \sin\lambda), \qquad p = \frac{q}{(q/p)}. \tag{1}$$

Why only the 3×3 block. The invariant mass depends on the momenta, not on where the tracks are: $\partial m/\partial d_{xy} = \partial m/\partial d_{sz} = 0$, so the last two columns of the Jacobian (Step 2) vanish. The ϕ–d_{xy} and λ–d_{sz} correlations are large in the 5×5 matrix, but they multiply a zero and drop out. The correlations inside the block ((q/p)–λ, (q/p)–ϕ, λ–ϕ) do enter and must be kept.

This is a first-order statement. Tracks bend between the reference point and the D_s vertex, so the opening angles at the vertex differ by a few mrad from those at the reference point. That is a shift of the mass value, identical in data and MC, not an uncertainty. It is the piece the vertex fit would have brought in, together with its mismodelling.

Why the tracks are uncorrelated. Each track is fitted from its own hits, whose measurement errors are independent. The three 5×5 matrices are therefore independent blocks of one 15×15 matrix, with zeros between them. Two effects break this, and neither belongs in a per-event σ:

- shared hits between close tracks: rare, and the K^+K^- from a ϕ open up enough for this not to matter;
- common systematics (magnetic-field scale, alignment, material), which move all tracks of an event together: scale effects, corrected by the calibration, not random per-event spread.

After a vertex fit the refitted momenta become correlated, because they share one vertex; that is why the fitted covariance has cross terms. Since we do not use the vertex fit, the covariance is block-diagonal and the propagation is a sum over tracks.

3 Step 2: propagation

To first order each track contributes $J_i C_i J_i^T$, where J_i is the gradient of the mass with respect to that track's three helix parameters; the contributions add because the tracks are independent:

$$\sigma^2(m) = \sum_{i \in \{K_1, K_2, \pi\}} J_i \, C_i \, J_i^T, \qquad J_i = \left(\frac{\partial m}{\partial (q/p)_i}, \, \frac{\partial m}{\partial \lambda_i}, \, \frac{\partial m}{\partial \phi_i} \right), \tag{2}$$

with C_i the 3×3 block of Step 1. The chain rule splits J_i into two factors: the mass with respect to the momentum vector, and the momentum vector with respect to the helix parameters.

Mass with respect to one track's momentum. With $E = \sum_i E_i$, $\vec{P} = \sum_i \vec{p}_i$ and $E_i = \sqrt{p_i^2 + m_i^2}$, differentiating $m^2 = E^2 - |\vec{P}|^2$ gives

$$\vec{g}_i \equiv \frac{\partial m}{\partial \vec{p}_i} = \frac{1}{m} \left(E \, \frac{\vec{p}_i}{E_i} - \vec{P} \right) = \gamma \, (\vec{\beta}_i - \vec{\beta}), \tag{3}$$

where $\vec{\beta}_i = \vec{p}_i / E_i$ is the velocity of track i, and $\vec{\beta} = \vec{P}/E$, $\gamma = E/m$ belong to the $KK\pi$ system. The sensitivity of the mass to a track is the D_s boost times the track's velocity relative to the D_s. Two limits:

- D_s at rest: $\vec{g}_i = \vec{\beta}_i$. Only the momentum magnitude matters, $\partial m / \partial |\vec{p}_i| = \beta_i$, and a small rotation of one track leaves m unchanged at first order.
- Boosted D_s: $\vec{g}_i$ acquires a large component transverse to the track, and direction errors start to matter (the opening-angle term).

Momentum vector with respect to the helix parameters. For $|q| = 1$, from $p = q/(q/p)$ and Eq. (1),

$$\frac{\partial \vec{p}}{\partial (q/p)} = -q\, p\, \vec{p}, \qquad \frac{\partial \vec{p}}{\partial \lambda} = p\, \hat{e}_\lambda, \qquad \frac{\partial \vec{p}}{\partial \phi} = p_T\, \hat{e}_\phi, \tag{4}$$

with the unit vectors

$$\hat{e}_\lambda = (-\sin\lambda\cos\phi,\ -\sin\lambda\sin\phi,\ \cos\lambda), \qquad \hat{e}_\phi = (-\sin\phi,\ \cos\phi,\ 0). \tag{5}$$

(The first follows from $\mathrm{d}p/\mathrm{d}(q/p) = -q/(q/p)^2 = -q\,p^2$ and $\vec{p} = p\,\hat{p}$.)

The Jacobian. Projecting each column on $\vec{g}_i$:

$$J_i = \Big(-q_i\, p_i\, (\vec{g}_i \cdot \vec{p}_i),\ \ p_i\, (\vec{g}_i \cdot \hat{e}_{\lambda,i}),\ \ p_{T,i}\, (\vec{g}_i \cdot \hat{e}_{\phi,i}) \Big). \tag{6}$$

Reading each term. Since $\sigma(q/p)/|(q/p)| = \sigma_p/p$, the (q/p) term of track i alone gives

$$\delta m_i\big|_p = (\vec{g}_i \cdot \hat{p}_i)\ \sigma_{p,i}, \tag{7}$$

the component of the relative velocity along the track times the momentum error. The two angle terms give $p_i\, (\vec{g}_i \cdot \hat{e}_{\lambda,i})\, \sigma_{\lambda,i}$ and $p_{T,i}\, (\vec{g}_i \cdot \hat{e}_{\phi,i})\, \sigma_{\phi,i}$. All three are combined through the full C_i, correlations included, not simply added in quadrature.

Where to evaluate J. Use the calibrated momenta (`pt`, `eta`, `phi`) with the kaon and pion mass hypotheses, and the measured $m_{KK\pi}$ in $1/m$: these are the momenta that define the mass being categorised. The calibration does not change directions and rescales $|\vec{p}|$ by a factor f close to 1, so J moves only at second order. Strictly, C_i refers to the uncalibrated (q/p), so its (q/p) row and column scale by $1/f$; a per-mille effect, but free to apply.

4 Step 3: the calibration term

The reco covariance describes the tracker's resolution as simulated. The calibration exists because the resolution in data is worse, so the missing width is added to each track's (q/p) variance. It is taken from the calibration itself, as an expected width rather than the smear actually applied.

What the calibration does. For every MC track, `TrackMomentumCalib` draws a standard normal number and rescales the momentum, leaving the direction unchanged:

$$z = \mathrm{gauss}(\mathrm{run}, \mathrm{lumi}, \mathrm{event}, \mathrm{track\ key}), \qquad f = \frac{p'}{p} = \mathrm{factor}(p, p_T, \eta, q, z).$$

f has two parts:

- a scale part, its value at $z = 0$, which shifts the momentum;
- a resolution part, its spread over z, which widens it.

Only the second is a resolution. The hadron terms apply (`kind='hadron'`); the muon terms are not needed for $m_{KK\pi}$.

The extra width, without z. For the track's own (p, p_T, η, q) define

$$\sigma_{\mathrm{cal},i} = \frac{f(z=+1) - f(z=-1)}{2\, f(z=0)}, \tag{8}$$

a relative width on p, and equally on (q/p) since $\sigma(q/p)/|(q/p)| = \sigma_p/p$ at first order. If f is not linear in z, use the RMS of $f/f(0)$ over $z \sim \mathcal{N}(0,1)$ instead; $f(\pm 2)$ tells whether that matters. Then

$$C_i[(q/p),(q/p)] \;\longrightarrow\; C_i[(q/p),(q/p)] + \left(\sigma_{\mathrm{cal},i}\,(q/p)_i\right)^2. \tag{9}$$

In the mass this adds $\sigma_{\mathrm{cal},i}\; p_i\,(\vec{g}_i \cdot \hat{p}_i)$ in quadrature for each track: the momentum term of Step 2 with σ_p/p replaced by σ_{cal}.

Why in quadrature. The calibration smear is drawn independently of the tracker measurement: it uses its own z and does not touch the direction, so it is uncorrelated with the angles. The z of different tracks are independent, so the term also adds track by track.

Why without z. σ_m must be a width: the expected mass error given the measured kinematics, not the error that actually occurred. The realised $f(z)$ of a given MC track is MC truth, since it says which way that track was smeared. Putting it into σ_m would correlate σ_m with the actual mass residual in MC, which data could never reproduce. Data has no z at all.

One assumption. This presumes that the reco covariance already describes the resolution of uncalibrated MC, so that adding σ_{cal} gives the resolution of calibrated MC, hence of data. The MC pull test of Section 6 (check 2) tests exactly this. If it fails, the reco covariance needs its own correction before anything is added.

Systematic variations. Build the categories with the nominal calibration. The calibration variations then move the resolution inside each category, a systematic on the per-category mass shapes; they should not move the category boundaries.

5 Step 4: one definition for data and MC

σ_m is a valid category variable only if it is the same function of reconstructed quantities in data and MC. Then calibrated MC, which reproduces the kinematics and resolution of data, also reproduces how events spread over σ_m and the mass resolution inside each category.

ingredient	data	MC
helix covariance C_i	packed candidate	packed candidate, same algorithm and format
momenta for J_i	measured	calibrated (`pt`, `eta`, `phi`), not `pt_raw`
$\sigma_{\mathrm{cal},i}$	from the track's (p, p_T, η, q), nominal	same function, same inputs
z, realised f, generator info	do not exist	exist, must not be used

Why data gets σ_{cal} too, although data is never smeared. The tracker computes C_i from its own error model, identically for data and simulation. In MC that model describes the simulated resolution. In data it describes the same simulated-quality resolution, which is smaller than the true one by exactly what the calibration adds to MC. Hence $C_i \oplus \sigma_{\mathrm{cal}}$ is the best estimate of the true width in both samples. Without σ_{cal} in data, its σ_m would be biased low and the per-category widths would no longer match calibrated MC.

What consistency buys, and what it does not.

- **Same meaning:** an event at a given σ_m has the same expected resolution in data and MC. This follows from the definition, provided the assumption of Step 3 holds.
- **Same population:** the σ_m distribution, hence the category fractions, agree between data and MC. This does *not* follow from the definition: it depends on how well MC models the track kinematics and the C_i themselves (hit patterns, η and p_T spectra), and must be measured (Section 6).

Ways to break it without noticing.

- using the vertex-fit covariance;
- using z or `calib_f` in the σ_{cal} term;
- using `pt_raw` in MC but measured momenta in data (different p in J and in the calibration bin);
- computing σ_{cal} in MC only;
- a covariance rescaling applied to one sample only: the branches `<trk>_cov_scale_*` record any `-cov-scale` factors; check that the stored covariance is the unscaled one, or divide the factors out, before using it in either sample.

A cheaper variant. Leaving σ_{cal} out in both samples is still a consistent definition, only a worse estimator: its pulls are wider than 1. If the categories are quantile cuts and σ_{cal} is a roughly constant fraction of the total, the ranking hardly changes. Where σ_{cal} varies strongly between bins (barrel vs endcap, low vs high p) it reorders events, so including it is the better default.

6 Validation

Five checks, in order, each testing a different assumption above. The first two can run on the current, uncalibrated, signal MC.

1. **Jacobian, numerically.** For a few hundred candidates move each helix parameter by a small step, recompute $m_{KK\pi}$, and compare the finite difference with Eq. (6). This catches sign and convention errors in the (q/p) column and in λ vs θ.
2. **MC pull, uncalibrated sample, without σ_{cal}.** On truth-matched $D_s \to \phi\pi$, form $(m_{KK\pi} - m_{\text{gen}})/\sigma_m$.
   - Compute m_{gen} from the generator momenta of the matched tracks (branches `gen_vis_{kp,km,pi}_*`): this removes FSR from the comparison; the D_s natural width is negligible.
   - Measure the core width (Gaussian fit to the core, or IQR/1.349): radiative tails are not part of σ_m.
   - Expect width 1, flat in quantiles of σ_m, in η and in p. This tests the assumption of Step 3.
   - A width different from 1 but constant is a global scale factor; one that varies across σ_m quantiles points to an error in C_i or J_i.
3. **MC pull, calibrated sample, with σ_{cal}.** Same test, width 1 expected. This checks the quadrature sum of Step 3 against the smear actually applied.

4. **D_s peak in data vs calibrated MC, per category** — the test that matters for the analysis.
   - Split the candidates in quantiles of σ_m (the categories, or finer).
   - In each, fit the $m_{KK\pi}$ peak with a free signal width and a linear or exponential background.
   - Plot the fitted width against $\langle\sigma_m\rangle$ in the bin: data and calibrated MC should lie on the same line, ideally of slope 1 through the origin.
   - Quote the per-category data/MC width ratio. It should be 1 within its error; otherwise it becomes a per-category resolution correction.
5. **Population.** Compare the σ_m distribution of D_s signal in data (sideband-subtracted or sPlot) with calibrated MC. The category fractions must agree. A residual difference means the MC kinematics or covariance modelling is off: reweight, or assign a systematic on the category yields. It says nothing about the σ_m definition.

Checks 4 and 5 use the $D_s \to \phi\pi$ peak itself, abundant in data and the same particle as in the signal: no control channel is needed.

7 Implementation with the existing branches

Everything needed is in the current ntuples except the calibration applier. A numpy sketch, vectorised over candidates:

```
import numpy as np

MASS = {'k1': 0.493677, 'k2': 0.493677, 'pi': 0.13957039}
PARS = ('qoverp', 'lambda', 'phi')          # first three helix parameters

def cov3(d, t):
    '''3x3 (q/p, lambda, phi) block; the branches are the upper triangle.'''
    n = len(d[t + '_pt'])
    C = np.empty((n, 3, 3))
    for a in range(3):
        for b in range(a, 3):
            C[:, a, b] = C[:, b, a] = d['%s_cov_%s_%s' % (t, PARS[a], PARS[b])]
    f = np.nan_to_num(d[t + '_calib_f'].astype(float), nan=1.0)   # data / uncalibrated MC: 1
    C[:, 0, :] /= f[:, None]                # q/p' = (q/p)/f
    C[:, :, 0] /= f[:, None]
    return C

def sigma_m_kkpi(d, sigma_cal=None):
    '''m(KKpi) and its per-candidate uncertainty. sigma_cal: {track: relative
    calibration width per candidate}, or None to leave the term out.'''
    trk = {}
    for t, mt in MASS.items():
        pt, eta, phi = (d['%s_%s' % (t, q)].astype(float) for q in ('pt', 'eta', 'phi'))
        p3 = np.stack([pt * np.cos(phi), pt * np.sin(phi), pt * np.sinh(eta)], 1)
        p = pt * np.cosh(eta)
        trk[t] = (p3, np.sqrt(p ** 2 + mt ** 2), p, pt, eta, phi)
    P = sum(v[0] for v in trk.values())
    E = sum(v[1] for v in trk.values())
    m = np.sqrt(E ** 2 - np.sum(P ** 2, 1))
    var = np.zeros(len(m))
    for t, (p3, Ei, p, pt, eta, phi) in trk.items():
        g = (E[:, None] * p3 / Ei[:, None] - P) / m[:, None]          # dm/dp_vec
        q = d[t + '_charge'].astype(float)
        sl, cl = np.tanh(eta), 1.0 / np.cosh(eta)                       # sin, cos lambda
        e_lam = np.stack([-sl * np.cos(phi), -sl * np.sin(phi), cl], 1)
```

```
        e_phi = np.stack([-np.sin(phi), np.cos(phi), np.zeros_like(phi)], 1)
        J = np.stack([-q * p * np.sum(g * p3, 1),                    # d/d(q/p)
                      p * np.sum(g * e_lam, 1),                      # d/d(lambda)
                      pt * np.sum(g * e_phi, 1)], 1)                 # d/d(phi)
        C = cov3(d, t)
        if sigma_cal is not None:
            C[:, 0, 0] += (sigma_cal[t] / p) ** 2                    # (sigma_cal * q/p)^2
        var += np.einsum('ni,nij,nj->n', J, C, J)
    return m, np.sqrt(var)
```

The calibration width, once `rds_trkcalib.py` and its JSON are available (`TrackCalib.factor` as called in `TrackMomentumCalib.z_and_f`; wrap it in `np.vectorize` if it takes scalars):

```
def sigma_cal_rel(cal, p, pt, eta, q, kind='hadron', variation='nominal'):
    f = lambda z: cal.factor(p, pt, eta, q, z, variation, kind=kind)
    return (f(1.0) - f(-1.0)) / (2.0 * f(0.0))
```

Notes.

- Feed the calibrated momenta to the bin lookup in MC and the measured ones in data. The calibration was defined on uncalibrated momenta, but f is within a per mille of 1, so this matters only at bin edges, and it keeps the definition symmetric.
- Skip or flag candidates with `<trk>_cov_pos_def` false; check what `<trk>_cov_fixed` means for them before trusting their covariance.
- Packed-candidate covariances are stored with reduced precision: fine for a per-event σ, and check 2 would show it otherwise.
- For the next production the same function can fill a convenience branch (`kkpi_mass_err`), so the analysis does not need the applier.

8 Check yourself

1. A D_s is produced at rest and decays to $K^+K^-\pi^+$. Which of the three helix-parameter errors drive σ_m, and why do the other two drop out at first order?
2. Boost the same D_s to 15 GeV. Using $\vec{g}_i = \gamma(\vec{\beta}_i - \vec{\beta})$, predict which tracks dominate σ_m: the slow or the fast ones, those at large or small angle to the D_s?
3. The ϕ–d_{xy} correlation in the 5×5 covariance is strong for low-p_T tracks. Does it change σ_m? What would change if $m_{KK\pi}$ were computed with the momenta at the D_s vertex instead?
4. A colleague adds σ_{cal} to MC only, "because only MC was smeared". How do the per-category D_s peak widths in data and MC now disagree, and in which direction?
5. In check 2 the pull width is 1.15 in the lowest σ_m quantile and 1.00 in the highest. Which ingredient do you suspect first, and which plot tells you?